Glen A. Dsouza

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Engineering Portfolio → flame1113.github.io/Portfolio (project write-ups, 3D models, schematics & build documentation)

EDUCATION

California Polytechnic State University, San Luis Obispo — Expected May 2030
B.S. Electrical Engineering — Incoming, College of Engineering

Newbury Park High School, Newbury Park, CA — Expected June 2026
GPA: 4.00 UW / 4.71 W

TECHNICAL PROJECTS

Power Wheels — Centralized Vehicle Control Unit (CVCU) — 2024 — 2025
Designer & Integrator

- Designed a custom PCB as the centralized control hub for a modified Power Wheels ride-on, integrating ESP32-WROOM-32U telemetry, RFID access control, OLED UI, and DFR0299 audio amplification
- Implemented AO3400 N-channel MOSFET propulsion switching with SPDT relay isolation and heavy-duty motor terminals; populated the board via reflow and precision soldering

Custom Mini-Z 1:28 RC Platform — July 2025 — Oct 2025
Designer & Fabricator

- Co-designed a 1:28 scale RC platform pairing a custom KiCad controller PCB (ATtiny85, DRV8833PWP) with a 3D-printed chassis under sub-millimeter packaging constraints
- Integrated propulsion and steering servo subsystems; independently assembled, soldered, and validated electromechanical performance

Numeric Operated Door Lock — May 2024 — July 2024
Project Developer

- Built an ESP32-based access system with helix keypad input, 5 V servo actuation, LCD feedback, and buzzer status signaling
- Developed control firmware in Arduino C++ for password validation and lock state management

EXPERIENCE

Kumon — Thousand Oaks, CA — May 2024 — Present
Paid Tutor

- Evaluate and grade math and reading assessments for students across grade levels
- Deliver targeted instructional support to students requiring additional concept reinforcement

SparkPrintz — Remote — June 2024 — Aug 2024
Sales & Marketing Intern

- Contacted school administrators and faculty to market SparkPrintz apparel printing services
- Conducted market research and authored business analysis write-ups on company positioning

TECHNICAL SKILLS

Programming: C++, Python, Java, JavaScript, HTML/CSS, SQL

EE / Design: KiCad, Altium, AutoCAD, PCB Layout, SMD Assembly, Power Electronics

Embedded: ESP32, ATtiny85, Arduino, PWM Motor Control, I^2C, UART

CAD / Fab: Fusion 360, FDM 3D Printing, Reflow Soldering, Enclosure Design